long time compared to that of a nuclear rotation, which is about 10^{-23} s. Consequently the wave mechanic properties of particles (and γ-rays) introduce a certain fundamental uncertainty in the particle energy. Several applications of this are described in later chapters.

5.19. EXERCISES

For some of the problems necessary nuclear data are given in the Tables or appendices.

5.1. ^{239}Pu emits α-particles of maximum 5.152 MeV. What is the recoil energy of the product formed?

5.2. Using a magnetic spectrometer the maximum energy of the electrons from ^{137}Cs was found in Figure 5.1 to correspond to 3.15×10^{-3} Tesla m. Calculate the energy (a) assuming that the electrons are non-relativistic, (b) with correction for relativistic mass increase.

5.3. ^{11}C decays through emission of positrons of a maximum energy of 1.0 MeV. Calculate the recoil energy of the daughter.

5.4. ^{16}N decays through β^- decay to ^{16}O with a half-life of 7.1 s. A number of very energetic γ's follow after the β-emission, the dominating one with an energy of 6.14 MeV. What is the ^{16}O recoil energy?

5.5. The binding energy of a K-electron in barium is 37 441 eV. Calculate from Figure 5.1 the internal conversion energy for ^{137m}Ba (Fig. 5.5).

5.6. From the specific activity of potassium (1850 dpm/g K) and the fact that it all originates in the rare isotope ^{40}K (0.0117%), calculate the half-life of ^{40}K.

5.7. One may assume that when ^{238}U was formed at the genesis an equal amount of ^{235}U was formed. Today the amount of ^{238}U is 138 times the amount of ^{235}U. How long a time ago did the genesis occur according to this assumption?

5.8. The interior of the earth is assumed to be built up of a solid core (radius 1371 km) followed by a molten core (radius 3471 km) and a molten mantle (radius 6354 km) covered by a 17 km thick crust. One assumes that 2% by weight of the molten mantle and crust is potassium; the average mantle density is assumed to be 6000 kg m^{-3} and that of the crust 3300 kg m^{-3}. What energy outflow will the radioactive decay of this element cause at the earth's surface? The decay scheme of ^{40}K is given in Eqn. (13.7). For the EC branch $Q = 1.505$ MeV, for the β^- branch 1.312 MeV. Each decay by the EC branch leads to emission of a 1.46 MeV γ. Compare this energy output to the solar energy influx to the earth of 3.2×10^{24} J y^{-1}.

5.9. A hospital has a 1.5 Ci source of ^{226}Ra in the form of a $RaBr_2$ solution. If the ^{222}Rn is pumped out each 48 h, what is (a) the radon activity (in Bq) at that moment, (b) the radon gas volume at STP?

5.10. (a) Prove the correctness of Eqn. (5.25) by using Newton's laws of motion. (b) Prove the correctness of Eqn. (5.48).

5.11. A recently prepared ^{212}Pb sample has the activity of 10^6 dpm. (a) What is the activity 2 h later? (b) How many lead atoms are left in the sample at this moment? $t_{½}$ 10.64 h.

5.12. A radioactive sample was measured at different time intervals:

Time (h)	Count rate (cpm)	Time (h)	Count rate (cpm)
0.3	11100	30	1015
5	5870	40	888
10	3240	50	826
15	2005	100	625
20	1440		

Determine the half-lives of the two nuclides (not genetically related) in the sample and their activities (in Bq) at time $t = 0$. The background of the detection device was 100 376 counts per 1 000 min; its counting efficiency was 17%.

5.13. The α-activity of a mixture of astatine isotopes was measured at different times after their separation, giving the following results:

t(min)	A(dpm)	t(min)	A(dpm)
12	756	121	256
17.2	725	140	215.5
23.1	638	161	178.5
30.0	600	184	150.7
37.7	545	211	127.3
47.5	494	243	101.9
59.5	435	276	84.9
73	380	308	68.2
87	341	340	55.0
102	288		

Calculate the half-lives of the isotopes and their activities at $t = 0$.

5.14. In the ion source of a mass spectrograph, UF_6 vapor is introduced which partly becomes ionized to UF_5^+. The ionic currents were measured at mass positions 333, 330, and 329. The ion current ratios were $I_{333}/I_{330} = 139$, and $I_{330}/I_{329} = 141.5$. What is the half-life of ^{234}U if that of ^{238}U is 4.5×10^9 y? Radioactive equilibrium is assumed to exist in the UF_6.

5.20. LITERATURE

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